Lean Canvas: Business Architecture

Validated B2B2C economic engine and distribution model for visual-to-tactile assistive tech.

1. PROBLEM

- Severe spatial isolation (285M visually impaired).
- Audio fatigue & ear blockage from talking screen readers (masks critical traffic sounds).
- Clunky traditional canes fail on head-level hazards.

EXISTING ALTERNATIVES:

White cane ($30), Guide dog ($40k+), 7Sense SuperBrain ($12,000+).

2. SOLUTION

- Dual wide-angle stereo vision & depth sensors.
- Real-time on-device edge AI (<100ms obstacle detection & depth mapping).
- 5×5 abdominal vibrotactile haptic matrix conveying direction, proximity, and hazard level silently.

3. VALUE PROPOSITION

"Spatial Sight Through Touch": Real-time 3D spatial awareness translated into silent tactile feedback.

HIGH-LEVEL CONCEPT:

"Autonomous Spatial Awareness for the Blind — Ear-Free, Hands-Free."

4. UNFAIR ADVANTAGE

- Ear-Free safety patent-pending tactile mapping architecture.
- 95%+ lower cost compared to European competitor 7Sense ($300 vs $12,000).
- Proprietary low-latency edge tactile encoding algorithms.

5. CUSTOMER SEGMENTS

- Primary: Blind & severely visually impaired individuals (40M worldwide).
- Institutional Buyers: Rehabilitation NGOs, Special Education institutions, Ministry of Social Solidarity.
- Early Adopters: Tech-savvy blind professionals and university students.

6. KEY METRICS

- Obstacle Avoidance Success Rate (>95%).
- User Spatial Orientation Speed (+40% faster navigation).
- System End-to-End Latency (<100ms benchmark).
- Device Adoption & Monthly Active Usage rate.

7. CHANNELS

- B2B partnerships with blindness associations (e.g., Al Nour Wal Amal, Resala, Baseera).
- Direct-to-Consumer e-commerce & assistive tech hubs.
- Government disability subsidy & medical device inclusion programs.

8. COST STRUCTURE

- Bill of Materials (BOM): ERM/LRA vibration actuators, edge MCU/SoC, camera modules (~$85/unit at volume).
- Manufacturing & Assembly (local fabrication & testing).
- Edge AI inference optimization & R&D.

9. REVENUE STREAMS

- Hardware Sales: Core Haptic Belt + Vision Sensor kit ($249 – $349 target retail, subsidized via NGOs).
- SaaS Navigation Upgrade: Cloud mapping, indoor spatial updates, custom AI hazard detection ($9.99/mo).
- B2B Training & Maintenance support contracts.